UNIVERSITY OF LONDON

MSci/MSc EXAMINATION 2024

For Internal Students of
Royal Holloway

DO NOT TURN OVER UNTIL TOLD TO BEGIN

CS4100/CS5100: Data Analysis

Time Allowed: **TWO hours**

Please answer **ALL** questions

EPMS calculators are permitted

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1. **Linear Regression** (25 marks)

   (a) Explain the difference between the label spaces of classification and regression problems. Why would you use a linear regression model for an inference task? Include a real-world example in your explanation. [5 marks]

   (b) Write the mathematical definition of a linear regression model, $f(X) \in \mathbb{R}$, when the input is $X \in \mathbb{R}$ and the model parameter is $\lambda \in \mathbb{R}^2$. [5 marks]

   (c) How many degrees of freedom does $f(X)$ have when $X \in \mathbb{R}^d$? Discuss the relationship between the number of degrees of freedom, the size of the model parameter, λ, the dimensionality of the input, d, and the model flexibility. [5 marks]

   (d) Write the formula of the Residual Sum Of Squares for a linear regression model, $f(X) \in \mathbb{R}$, on a given data set, $D = \{(X_n, Y_n) \in \mathbb{R} \times \mathbb{R}\}_{n=1}^{N}$. [5 marks]

   (e) What is a polynomial regression model? Give an explicit example and explain why you can say that the model is *nonlinear* in the input, X, but *linear* in the parameter, λ. You can assume $X \in \mathbb{R}$. [5 marks]

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2. **Decision Trees (25 marks)**

(a) Explain what a Decision Tree is. How can you use it in a data analysis task? What is the difference between Classification and Regression Decision Trees? [5 marks]

(b) Imagine you have a trained Decision Tree, T. Describe how you can use T to predict the label of a test input, X. Why can you say that T is equivalent to a partition on the input space? [5 marks]

(c) A Decision Stump, T_1, is a Decision Tree with one root and two terminal nodes. Assume the object attributes are one-dimensional, i.e., $X \in \mathbb{R}$. Why do you need a single real-valued threshold, ξ, to define T_1? Let $\xi = 2$ and draw a diagram of the tree and the corresponding partition of the input space. [5 marks]

(d) Assume you found $\xi = 2$ by training T_1 on

$$\begin{aligned}\mathcal{D} &= \{(x_n, y_n) \in \mathbb{R} \times \{\texttt{yes}, \texttt{no}\}\}_{n=1}^{N} \\ &= \{(-1, \texttt{no}), (1, \texttt{no}), (4, \texttt{yes}), (10, \texttt{no}), (13, \texttt{no})\}\end{aligned}$$

What label does the tree predict on input $X = 5$? Explain how you produce the prediction using the tree output and a Bayes classifier. [5 marks]

(e) Let B_1 and B_2 be the sets of training objects, $(x_n, y_n) \in \mathcal{D}$, assigned to the terminal nodes of T_1. The unnormalized total entropy of (the partition associated with) T_1 is $H^*(T_1) = |B_1|H(B_1) + |B_2|H(B_2)$, where $|B_i|$ is the size of B_i, $i = 1, 2$. Compute $H^*(T_1)$ (let $\xi = 2$).

Hint: In a binary classification problem, the entropy of a set, D, is

$$\begin{aligned}H(D) &= -p_{\texttt{yes}} \log(p_{\texttt{yes}}) - p_{\texttt{no}} \log(p_{\texttt{no}}) \\ &= -p_{\texttt{yes}} \log(p_{\texttt{yes}}) - (1 - p_{\texttt{yes}}) \log(1 - p_{\texttt{yes}})\end{aligned}$$

where $p_{\texttt{yes}}$ is the proportion of positive labels in D, i.e.,

$$p_{\texttt{yes}} = N^{-1} \sum_{n=1}^{N} \mathbf{1}(Y_n = \texttt{yes}).$$

You do not need to evaluate the logarithms in your final expression. [5 marks]

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3. **Multiple testing (23 marks)**

(a) Give a real-life example illustrating the need for correction for multiple hypothesis testing. [4 marks]

(b) Define the FWER (family-wise error rate). [4 marks]

(c) Suppose you have tested 5 new drugs, D_1, D_2, D_3, D_4, and D_5. Formally, for each drug D_j we tested the null hypothesis that the new drug is not better than the standard drug and obtained a p-value of p_j. The resulting p-values are

drug	p_1	p_2	p_3	p_4	p_5
p-value	0.01	0.01	0.015	0.02	0.4

Assume that the significance level is 0.05. Explain which one of the 5 null hypotheses you will reject when using:

i. The Bonferroni method. [4 marks]

ii. The Holm method. [6 marks]

Hint: The main formula of the Holm method is

$$L := \min \left\{ j : p_{(j)} > \frac{\alpha}{m+1-j} \right\}$$

(using the module's notation).

(d) Which method should you use in this case? Discuss the pros and cons of both methods. [5 marks]

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4. **Page Rank (27 marks)**

(a) Define the PageRank algorithm for ranking web pages and briefly explain the intuition behind it. [9 marks]

Hint: The main formula of the PageRank algorithm is

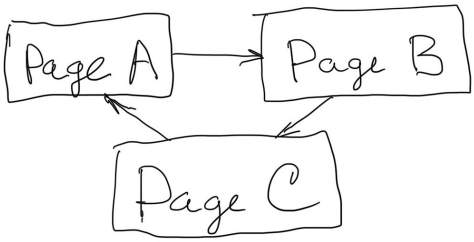
$$p_i = (1-d) + d\sum_{j=1}^{N} \frac{L_{ij}}{c_j} p_j,$$

but you should explain the notation and the steps of the algorithm.

(b) What is the difference between the simplified and proper versions of the PageRank algorithm? [4 marks]

(c) Briefly discuss the role of the *damping factor* in the PageRank algorithm. [4 marks]

(d) Apply PageRank in this situation



In words: we consider three web pages, A, B, and C. Page A links to page B, page B links to page C, page C links to page A, and these are the only links present in the three documents. Assume a damping factor of 0.5, $d = 0.5$. You should compute the PageRanks of all three pages. [7 marks]

(e) Explain the result that you obtained without (explicitly) referring to the PageRank algorithm. [3 marks]

END

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